

LECTURE 22

Multi-Valued Dependencies and Fourth Normal Form

CSC270 · Database Systems

COMSATS University Islamabad, Lahore Campus

What We'll Cover

- Why BCNF alone does not catch every source of redundancy
- Multi-valued dependencies: notation $X \twoheadrightarrow Y$ and what it means
- Trivial vs. non-trivial MVDs, and the properties of MVDs
- Inference rules for MVDs, parallel to Armstrong's Axioms
- Fourth Normal Form: definition and test
- A complete worked example: Employee, Skills, and Languages
- 4NF decomposition and the lossless-join property

A Table With Two Independent Facts

EmpSkillsLangs (empNo, skill, language)

empNo	skill	language
E1	Java	English
E1	Java	Urdu
E1	Python	English
E1	Python	Urdu
E2	SQL	English

Two independent facts about each employee: the skills they have, and the languages they speak. Both are one-to-many, and they are stored together in the same table.

Why BCNF Cannot See the Problem

Check BCNF formally. Are there any non-trivial functional dependencies?

Candidate FD	Holds?	Why
$\text{empNo} \rightarrow \text{skill}$	No	E1 has two skill values (Java, Python)
$\text{skill} \rightarrow \text{language}$	No	Java goes with both English and Urdu
$\text{empNo} \rightarrow \text{language}$	No	E1 has two language values

So the relation is already in BCNF

With no non-trivial FDs, nothing can violate BCNF. The only candidate key is the full set {empNo, skill, language}, an all-key relation. BCNF has no way to see the redundancy in this table.

The Spurious Combination

E1's two skills and two languages were cross-multiplied into four rows.

Row 2 claims: E1's Java skill is paired with Urdu

Skill and language were never linked for E1. They are two unrelated facts forced into one row.

Row 3 claims: E1's Python skill is paired with English

This pairing is meaningless. The table just happens to store both combinations.

The real structure

Two independent one-to-many facts: E1 knows {Java, Python} and E1 speaks {English, Urdu}. The four E1 rows encode two independent lists of two values each (2 + 2 facts), not four correlated pairs.

The Update Anomaly

Now E1 learns French. How many rows must we insert?

empNo	skill	language
E1	Java	English
E1	Java	Urdu
E1	Python	English
E1	Python	Urdu
E1	Java	French
E1	Python	French

Two rows, not one

Inserting only (E1, Java, French) leaves “does E1 speak French?” with a different answer depending on which row you check. Forgetting a row is an update anomaly.

Multi-Valued Dependencies, Defined

A multi-valued dependency $X \twoheadrightarrow Y$ over R (with $Z = R - X - Y$) holds when, for each value of X , the set of Y -values is completely independent of the Z -values.

Formal test: tuples t_1 and t_2 agree on X

Then R must also contain a tuple with t_1 's X and Y values and t_2 's Z value, and one with t_2 's X and Y values and t_1 's Z value.

In EmpSkillsLangs: $\text{empNo} \twoheadrightarrow \text{skill}$

Here $X = \{\text{empNo}\}$, $Y = \{\text{skill}\}$, $Z = \{\text{language}\}$. Skills are independent of languages for each employee.

Complementation: if $\text{empNo} \twoheadrightarrow \text{skill}$ holds, so does $\text{empNo} \twoheadrightarrow \text{language}$.

Applying the Definition to the Table

Check empNo $\rightarrow$ skill with two tuples that agree on empNo = E1:

Tuple	empNo	skill	language	Role
t ₁	E1	Java	English	given
t ₂	E1	Python	Urdu	given
needed	E1	Java	Urdu	present ✓
needed	E1	Python	English	present ✓

Conclusion: the MVD holds

Both mixed tuples the definition requires are already in the table. The two skills combine with both languages, so skill is independent of language for E1. For E2 there is only one row, so the condition holds trivially.

The Intuitive Test

The intuitive test: for a fixed employee, does knowing which language they speak change which skills are possible?

1

Independent: an MVD

The two lists are just attached to the same employee. Knowing one tells you nothing new about the other. This is an MVD, and the cross-product is spurious.

2

Interacting: no MVD

If Java were certified in English and Python in Urdu, the skill-language link becomes real. The MVD breaks, and the cross-product is no longer spurious.

Trivial and Non-Trivial MVDs

1

Trivial MVD

$X \twoheadrightarrow Y$ is trivial if $Y \subseteq X$, or if $X \cup Y = R$. It holds in every relation and says nothing about design.

2

Non-trivial MVD

Neither condition holds. Only non-trivial MVDs can violate 4NF. $\text{empNo} \twoheadrightarrow \text{skill}$ is non-trivial: skill is not in empNo, and $\{\text{empNo}, \text{skill}\}$ misses language.

Properties of MVDs

- Every FD is also an MVD: if $X \rightarrow Y$, then for fixed X there is exactly one Y , so there is no choice to be independent about.
- MVDs always come in a complementary pair: $X \twoheadrightarrow Y$ over $\{X, Y, Z\}$ holds if and only if $X \twoheadrightarrow Z$ holds.
- Checking an MVD means looking at the whole relation, not one row pair. It says which combinations of rows must exist.

Deriving New MVDs

Rule	Statement
Complementation	If $X \twoheadrightarrow Y$ holds over R , then $X \twoheadrightarrow (R - X - Y)$ also holds.
Augmentation	If $X \twoheadrightarrow Y$ and $W \supseteq Z$, then $WX \twoheadrightarrow YZ$ holds.
Transitivity	If $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$, then $X \twoheadrightarrow (Z - Y)$ holds.
Replication	If $X \rightarrow Y$ holds, then $X \twoheadrightarrow Y$ holds (every FD is an MVD).
Coalescence	If $X \twoheadrightarrow Y$, and $W \cap Y = \emptyset$ with $W \rightarrow Z$ for some $Z \subseteq Y$, then $X \rightarrow Z$.

Why 4NF Exists Beyond BCNF

1

BCNF only sees FDs

BCNF tests functional dependencies. EmpSkillsLangs has none, so BCNF passes it with no warning.

2

MVDs need their own rule

Redundancy can come from independent multi-valued facts. A stronger normal form, 4NF, is needed to remove it. 4NF is to MVDs what BCNF is to FDs.

Fourth Normal Form (4NF)

A relation R is in Fourth Normal Form if, for every non-trivial MVD $X \twoheadrightarrow Y$ over R, X is a superkey of R.

Same shape as BCNF

Replace “functional dependency” with “multi-valued dependency” and the rule is identical.

Every FD is also an MVD

So 4NF automatically re-checks every BCNF condition, plus the MVD conditions BCNF cannot see.

4NF Is Strictly Stronger Than BCNF

BCNF

Every non-trivial FD has a determinant that is a superkey. EmpSkillsLangs passes: it has no FDs at all.

Looks only at functional dependencies.

4NF

Every non-trivial MVD has a superkey determinant.
EmpSkillsLangs fails: $\text{empNo} \twoheadrightarrow \text{skill}$ is non-trivial, and empNo is not a superkey.

Implies BCNF: every 4NF relation is also in BCNF.

Testing EmpSkillsLangs Against 4NF

Test	Result for EmpSkillsLangs
Candidate key	{empNo, skill, language}: the full set
Non-trivial MVD	empNo $\twoheadrightarrow$ skill (skill $\not\subseteq$ empNo; {empNo, skill} misses language)
Is empNo a superkey?	No: it does not even determine one skill value
4NF verdict	VIOLATES 4NF

The 4NF Split Rule

Given R violating 4NF via non-trivial $X \twoheadrightarrow Y$, with $Z = R - X - Y$, replace R with:

$$R_1 = X \cup Y$$

For our example: $R_1 = \{\text{empNo}\} \cup \{\text{skill}\} = \text{EmpSkills}(\text{empNo}, \text{skill})$

$$R_2 = X \cup Z$$

For our example: $R_2 = \{\text{empNo}\} \cup \{\text{language}\} = \text{EmpLanguages}(\text{empNo}, \text{language})$

X becomes the join column, the same role it played in the BCNF split.

The Two Resulting Tables

EmpSkills (empNo, skill)

empNo	skill
E1	Java
E1	Python
E2	SQL

EmpLanguages (empNo, language)

empNo	language
E1	English
E1	Urdu
E2	English

Adding a third skill for E1 costs one row

Before: a new skill meant one new row per language (2 rows). After: one new row in EmpSkills. Each table also records only its own fact, so no spurious pairing can be stored.

Redundancy Removed, Nothing Lost

Rows stored: the original table versus the decomposed pair

Design	Rows stored	Growth when E1 learns a 3rd skill
Original EmpSkillsLangs	5	multiplicative: +2 rows (one per language)
EmpSkills + EmpLanguages	$3 + 3 = 6$	additive: +1 row

The decomposed pair stores 6 rows, not fewer. Its advantage is that the cost of new facts grows additively, while the original table's cost grows multiplicatively with $|skills| \times |languages|$.

Checking Each Piece Is in 4NF

1

EmpSkills(empNo, skill)

Only key: {empNo, skill}, all attributes. Any MVD over a two-attribute relation is trivial, because $Y \subseteq X$ or $X \cup Y = R$. So the relation is in 4NF.

2

EmpLanguages(empNo, language)

Only key: {empNo, language}, all attributes. Same argument: no non-trivial MVD is possible, so the relation is in 4NF.

Lossless-Join Property, Confirmed

EmpSkills ⋈ EmpLanguages (on empNo)

empNo	skill	language
E1	Java	English
E1	Java	Urdu
E1	Python	English
E1	Python	Urdu
E2	SQL	English

Lossless by Fagin's theorem

R decomposes losslessly into XY and XZ if and only if $X \twoheadrightarrow Y$ holds over R. Here $\{\text{empNo}\} \twoheadrightarrow \{\text{skill}\}$ holds, so the join returns exactly the 5 original rows.

Key Takeaways

- A multi-valued dependency $X \twoheadrightarrow Y$ says that, for a fixed X , the Y -values are independent of every other attribute. Every FD is an MVD.
- MVDs come in complementary pairs: $X \twoheadrightarrow Y$ over $\{X, Y, Z\}$ if and only if $X \twoheadrightarrow Z$.
- 4NF requires every non-trivial MVD's determinant to be a superkey. It is the same rule as BCNF, one level stronger.
- EmpSkillsLangs is in BCNF but not 4NF. Splitting into EmpSkills and EmpLanguages removes the spurious cross-product, losslessly.
- 4NF removes physical redundancy and its update anomalies. It does not remove any information the join can express.